

A ROBUST TEST FOR THE ONE-SAMPLE DISPERSION PROBLEM BASED ON A U STATISTIC

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SUMMARY

A test, based on a U statistic which is the minimum variance unbiased estimator of a new dispersion measure, is proposed for the one-sample dispersion problem and it is shown that this test is generally preferable to the tests proposed by Crouse and Lemmer (1977).

1. Introduction and Main Results

For a random sample $X_1, \dots, X_n$ from a distribution with unknown, continuous c.d.f. $F(x)$ the number A defined by

$$P[(X_i - X_j)^2 > A^2] = p_0 = 0,10 \text{ for } i, j \in \{1, \dots, n\}, i \neq j$$

can be used as a general measure of dispersion or spread. (We shall not distinguish between these two concepts.)

Consider the null Hypothesis $H_0 : A = A_0$ and the alternative hypothesis $H_\theta : A = A_0(1 + \theta), \theta > 0$.

Defining $U_n = \binom{n}{2}^{-1} \sum_{i < j} I[(X_i - X_j)^2 > A_0^2]$ with $I(\cdot)$ the indicator function, it follows from the theory of U statistics that U_n is a minimum variance unbiased estimator of $P[(X_i - X_j)^2 > A_0^2]$ and is asymptotically normally distributed for $n \rightarrow \infty$ (Hoeffding, 1948. This follows from the finiteness of $I_{2\theta}$ defined below.) The test statistic

$$U = \{U_n - E(U_n | H_0)\} / \{\text{var}(U_n | H_0)\}^{1/2}$$

is studied under alternative distributions of the form

$H_\theta : F_\theta(x) = F_0(x/(1 + \theta)), \theta = a/\sqrt{n} > 0$ with F_0 the c.d.f. under H_0 . Note that $1 + \theta = \sigma_\theta/\sigma_0, \sigma_\theta$ the standard deviation of F_θ .

It follows readily that

$$\begin{aligned}
E(U_n | H_\theta) &= P[(X_i - X_j)^2 > A_0^2 | H_\theta] \\
&= \int_{-\infty}^{\infty} [1 - F_\theta(x + A_0) + F_\theta(x - A_0)] dF_\theta(x) \\
&= I_{1\theta}, \text{ say, and}
\end{aligned}$$

$$\text{var}(U_n | H_\theta) = [2 I_{1\theta} + 2(3 - 2n)I_{1\theta}^2 + 4(n - 2)I_{2\theta}] / n(n - 1) \text{ where}$$

$$I_{2\theta} = \int_{-\infty}^{\infty} [1 - F_\theta(x + A_0) + F_\theta(x - A_0)]^2 dF_\theta(x).$$

Hence $E(U_n | H_0) = p_0$ and

$$\text{var}(U_n | H_0) = [2p_0 + 2(3 - 2n)p_0^2 + 4(n - 2)I_{20}] / n(n - 1).$$

Reject H_0 if $U \geq z_\alpha$ with $z_\alpha = \Phi^{-1}(1 - \alpha)$.

The a.r.e. of this test with respect to the X^2 test is given by

$$A_0^2 \left(\int_{-\infty}^{\infty} f_0(x + A_0) f_0(x) dx \right)^2 (\mu_4 / \sigma_0^4 - 1) / \{4(I_{20} - p_0^2)\}$$

with $f_0(x) = \frac{d}{dx} F_0(x)$, μ_4 and σ_0^2 the central moments of the distribution with c.d.f. F_0 , assuming that $\frac{d}{dx} f_0(x)$ exists. In the case of normality this a.r.e. assumes its maximum value of presumably 1 (the value 0.993 has been obtained by quadrature) in the vicinity of $p_0 = 0.10$.

Since I_{20} depends on the type of distribution F (but is invariant to location and scale transformations — cf. Appendix), the utility of the test is limited — it is not distribution-free. However, a robust test with good power properties in comparison with other tests for the same problem can be constructed by using an estimate of I_{20} — cf. Table 1.

Since F_0 is unknown, so is the value of I_{20} , but it is easily shown that $p_0^2 \leq I_{20} \leq I_{2\theta}$ for the hypotheses considered. A generally conservative test can be constructed by using, say, $I_{20} = 0.037$ throughout. But for θ close to 0 (i.e. close to H_0), the estimation of I_{20} by utilizing an unbiased estimator of $I_{2\theta}$, namely

$$\hat{I}_{2\theta} = n^{-1} \binom{n-1}{2}^{-1} \sum_i \sum_{\substack{j < j' \\ (\neq i) (\neq i)}} \mathbb{I}[(X_i - X_j)^2 > A_0^2 \text{ and } (X_i - X_{j'})^2 > A_0^2],$$

naturally gives a better test.

The following procedure is proposed:

- (i) If $\hat{I}_{2\theta} > 0,037$, take $I_{20} = 0,037$
- (ii) If $\hat{I}_{2\theta} < p_0^2$, take $I_{20} = p_0^2$
- (iii) If $p_0^2 \leq \hat{I}_{2\theta} \leq 0,037$, take $I_{20} = \hat{I}_{2\theta}$.

The calculation of $\hat{I}_{2\theta}$ is very simple by using a graphic method — cf. Appendix.

2. Simulated Sampling Experiments

The robustness and power properties of the test are examined by means of a Monte Carlo study. Every run consisted of 4 000 samples of size n . Estimates of the standard deviations are also given.

The notoriously nonrobust χ^2 test and the computationally complicated jackknife test (cf. Miller, 1968) with subsample size $k = 1$ have also been incorporated in the study — cf. Table 2. Empirical levels below α are not considered to be as serious as levels above α because in such cases the tests are conservative. The term “nonrobust” will mainly be associated with the latter case.

The χ^2 test has the highest power in most cases, but it is very nonrobust for non-normal distributions. The U test is always better than C_2 (cf. Crouse and Lemmer, 1977) and in general also better than the jackknife test, especially in the case of nonsymmetrical distributions. For symmetrical distributions there is not much to choose between U and C_3 (cf. Crouse and Lemmer, 1977), but in the case of non-symmetrical distributions U is much more robust than C_3 , which is in turn more robust than the χ^2 test.

TABLE 1

Values of I_{20} for different distributions; $p_0 = 0,10$.

Distribution	I_{20}	
Normal	0,024 (quadrature)	
Uniform	$2p_0^{3/2}/3$	= 0,021
Exponential	$(p_0 + p_0^2 + p_0^3)/3$	= 0,037
$\chi^2(4)$	0,033 (quadrature)	

TABLE 2

Empirical power ($\times 1\ 000$) of tests for samples of size n from different distributions, $\alpha = 0,05$. (Standard deviations ($\times 1\ 000$) given in brackets.)

Distribution	n	θ	TEST				
			U	C_2	C_3	J	χ^2
Normal	8	0	033(003)	006(001)	028(003)	031(003)	046(003)
		0,5	416(008)	174(006)	247(007)	352(008)	519(008)
		1	767(007)	454(008)	549(008)	687(007)	847(006)
	12	0	026(003)	013(002)	042(003)	036(003)	048(003)
		0,5	535(008)	231(007)	504(008)	529(008)	655(008)
		1	887(005)	552(008)	846(006)	880(005)	932(004)
	24	0	028(003)	013(002)	048(003)	046(003)	052(004)
		0,5	786(006)	466(008)	737(007)	830(006)	866(005)
		1	996(001)	898(005)	980(002)	993(001)	997(001)
Uniform	8	0	028(003)	005(001)	021(002)	045(003)	006(001)
		0,5	504(008)	359(008)	448(008)	499(008)	604(008)
		1	842(006)	662(007)	754(007)	811(006)	912(004)
	12	0	024(002)	009(001)	027(003)	049(003)	004(001)
		0,5	640(008)	464(008)	770(007)	737(007)	755(007)
		1	954(003)	804(006)	964(003)	954(003)	981(002)
	24	0	012(002)	008(001)	028(003)	053(004)	003(001)
		0,5	905(004)	843(006)	964(003)	959(003)	960(003)
		1	999(001)	991(001)	1000 -	1000 -	1000 -
Exponential	8	0	066(004)	030(003)	058(004)	027(003)	114(005)
		0,5	282(007)	180(006)	242(007)	178(006)	382(008)
		1	531(008)	359(008)	440(008)	386(008)	635(008)
	12	0	073(004)	034(003)	091(005)	030(003)	119(005)
		0,5	350(008)	209(006)	409(008)	259(007)	473(008)
		1	666(007)	465(008)	719(007)	552(008)	774(007)
	24	0	062(004)	036(003)	091(005)	034(003)	154(006)
		0,5	505(008)	369(008)	582(008)	454(008)	670(007)
		1	866(005)	745(007)	889(005)	841(006)	931(004)

TABLE 2 (Continued)

Distribution	n	θ	TEST				
			U	C_2	C_3	J	χ^2
$\chi^2(4)$	8	0	055(004)	021(002)	046(003)	025(002)	087(004)
		0,5	340(007)	180(006)	247(007)	224(007)	451(008)
		1	631(008)	407(008)	495(008)	512(008)	712(007)
	12	0	058(004)	025(002)	072(004)	029(003)	094(005)
		0,5	408(008)	223(007)	445(008)	325(007)	538(008)
		1	759(007)	492(008)	772(007)	695(007)	843(006)
	24	0	061(004)	030(003)	084(004)	026(003)	113(005)
		0,5	608(008)	402(008)	645(008)	563(008)	737(007)
		1	942(004)	825(006)	941(004)	932(004)	977(002)

APPENDIX

A1. STATEMENT: I_{20} depends on the type of distribution F , but is invariant to location and scale transformations.

Proof:

Let X have d.f. f_X and Z d.f. f_Z where $Z = (X - c)/d$, c and d constants.

If A is defined by $p_0 = \int_{-\infty}^{\infty} [1 - F_X(x + A) + F_X(x - A)]dF_X(x)$

and A' by $p_0 = \int_{-\infty}^{\infty} [1 - F_Z(z + A') + F_Z(z - A')]dF_Z(z)$,

the result is proved by showing that

$$\begin{aligned} & \int_{-\infty}^{\infty} [1 - F_X(x + A) + F_X(x - A)]^2 dF_X(x) \\ &= \int_{-\infty}^{\infty} [1 - F_Z(z + A') + F_Z(z - A')]^2 dF_Z(z). \end{aligned}$$

Using $F_Z(z) = F_X(dz + c)$, it follows that

$$\int_{-\infty}^{\infty} [1 - F_X(x + A) + F_X(x - A)]dF_X(x) = \int_{-\infty}^{\infty} [1 - F_Z(z + A') + F_Z(z - A')]dF_Z(z)$$

if $dA' = A$. Thus

$$\begin{aligned} & \int_{-\infty}^{\infty} [1 - F_Z(z + A') + F_Z(z - A')]^2 dF_Z(z) \\ &= \int_{-\infty}^{\infty} [1 - F_X(dz + dA' + c) + F_X(dz - dA' + c)]^2 dF_X(dz + c) \\ &= \int_{-\infty}^{\infty} [1 - F_X(x + A) + F_X(x - A)]^2 dF_X(x). \end{aligned}$$

A2 GRAPHIC CALCULATION OF $\hat{I}_{2\theta}$

Mark three points P, Q and R on a ruler such that $PQ = A_0$ units and $QR = A_0$ units. Plot all observations on a number axis, in these same units. Align the ruler on the number axis and place the centre point, Q, on the i -th observation. Count the number of observations, r_i , lying outside of the marked segment PR. Do this in turn for all observations. Then $I_{2\theta}$ may be computed as

$$\hat{I}_{2\theta} = n^{-1} (n-1)^{-1} \sum_{i=1}^n \binom{r_i}{2}.$$

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